

# A Multi-Modal Markov-Chain Passenger-Flow Model for the BKK Budapest GTFS Network

Entropy Maximisation · CTMC · Gillespie SSA · Network Resilience

Transport Modelling Working Group

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Code: `github.com/[repo]` · Data: `opendata.bkk.hu` (CC0-1.0)

# Outline

- 1 Motivation
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- 3 Entropy Maximisation
- 4 CTMC and Kolmogorov Equations
- 5 Gillespie Simulation
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# Why Model Passenger Flow Dynamics?

## BKK Budapest network

- $\approx 4 \times 10^6$  weekday journeys
- $\approx 5,500$  active stops
- 6 modes: bus, tram, trolleybus, metro, HÉV, ferry
- 368 routes, 178 083 trips/day

## Practical stakes

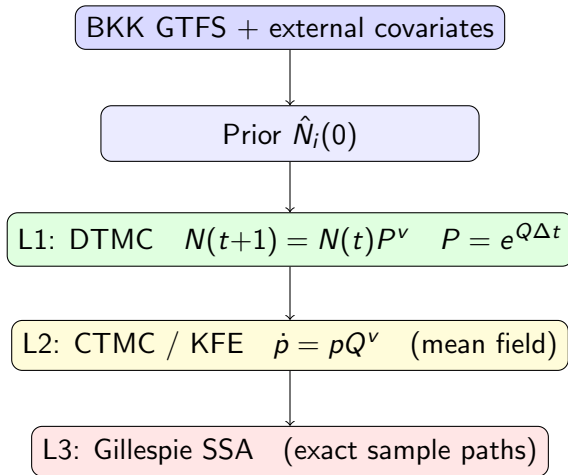
- Platform crowding risk
- Vehicle overcrowding
- Transfer delay propagation
- Disruption cascades
- Scenario / investment planning

**Key problem:** spatially resolved passenger density is not directly observable from the public schedule.

## Our approach

Derive a CTMC generator directly from the publicly available GTFS timetable + entropy-maximising trip distribution. No AFC/APC data required (but model is calibratable when they exist).

# Three Stochastic Layers



# GTFS Data Model

## Files used:

- stops.txt: vertex set  $S$ ,  
 $n \approx 5,551$
- routes.txt: mode  $v \in \mathcal{V}$
- stop\_times.txt: travel times  $\bar{\tau}_{ij}^v$
- shapes.txt: distances  $d_{ij}^v$
- transfers.txt: transfer penalties  
 $\delta_i$
- calendar\_dates.txt: weekday  
filter (BKK uses no calendar.txt)

## Network statistics (BKK):

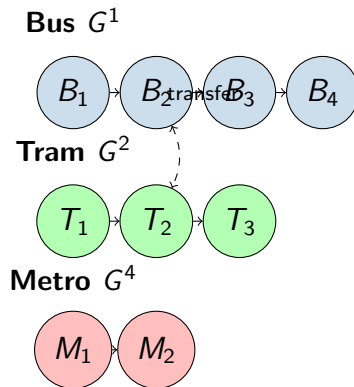
| Mode  | $ S^v $ | $ E^v $ |
|-------|---------|---------|
| Bus   | 4 151   | 17 824  |
| Tram  | 649     | 4 218   |
| Metro | 103     | 612     |
| Ferry | 2       | 2       |
| Total | 5 551   | 22 656  |

Bus fill ratio: 0.10% of all pairs  $\Rightarrow$  *sparse matrices mandatory*.

## Key design choice

Disconnected pairs are **absent** from the CSR structure, not assigned  $C^\infty$ . This prevents spurious small transition probabilities

# Modal Subgraph Structure



Modal subgraphs  $G^v = (S^v, E^v)$ ; inter-modal transfers handled via the penalty term in  $C_{ij}^v$ .

# Generalised Cost Matrix

For admissible pair  $(i, j) \in E^v$ :

$$C_{ij}^v = \underbrace{\kappa^v \bar{\tau}_{ij}^v}_{\text{in-vehicle [s]}} + \underbrace{\gamma d_{ij}^v}_{\text{distance [s]}} + \underbrace{\delta \mathbf{1}_{i \in \mathcal{T}} t_{\text{tf}}(i)}_{\text{transfer [s]}}$$

- $\kappa^v$ : mode comfort weight (metro: 0.40, bus: 1.00)
- $\gamma = 0.01 \text{ s m}^{-1}$ : distance disutility
- $t_{\text{tf}}(i)$ : min transfer time from `transfers.txt`

## Dimensional check

All terms in seconds  $\Rightarrow \lambda^v$  has units  $\text{s}^{-1} \Rightarrow \lambda^v C_{ij}^v$  dimensionless. ✓

**Disconnected pairs:** absent from CSR structure (not  $C^\infty = 10^6$ , which injects spurious transitions).

# Wilson Entropy Maximisation

**Primal problem:**

$$\max_{F_{ij}^v \geq 0} H = - \sum_{v,i,j} F_{ij}^v \ln F_{ij}^v$$

subject to:  $\sum_{j \in \mathcal{N}^v(i)} F_{ij}^v = N_i^v(0) \forall i, v$  (origin conservation) and  $\sum_{i,j} F_{ij}^v C_{ij}^v \leq \bar{C}^v$  (cost constraint).

Lagrangian optimality  $\partial \mathcal{L} / \partial F_{ij}^v = 0$  gives:

$$P_{ij}^v = \pi_{j|i}^v = \frac{\exp(-\lambda^v C_{ij}^v)}{\sum_{k \in \mathcal{N}^v(i)} \exp(-\lambda^v C_{ik}^v)}$$

$$\begin{array}{ccc} \lambda^v = 0 & \xrightarrow{\lambda^v \rightarrow \infty} & \lambda^v \\ \text{uniform} & & \text{shortest-path} \\ \text{distribution} & & \text{routing} \end{array}$$

# Self-Loops: A Common Modelling Pitfall

## Warning: self-loops in the DTMC

Several prior formulations set  $C_{ii}^v = 0$  and include the diagonal in the softmax denominator, yielding large  $P_{ii}^v \approx 0.8\text{--}0.95$  (“stay at stop”) — a modelling artefact, not a calibrated parameter.

### Our convention:

- DTMC step = one completed trip leg (board at  $i$ , alight at  $j$ ,  $j \neq i$ ).
- Softmax denominator runs over  $\mathcal{N}^v(i)$  only (no self-loop term).
- $P^v$  is the *embedded jump chain* of the CTMC (not uniformisation).
- “Waiting time” is absorbed into the holding-time distribution of the CTMC.

# From DTMC to CTMC: The Generator

**Key insight:** GTFS gives continuous departure times. Embed the jump chain in continuous time.

Definition (CTMC generator  $Q^\nu$ )

$$Q_{ij}^\nu = \begin{cases} q_{ij}^\nu = \mu_i^\nu \pi_{j|i}^\nu \geq 0 & j \neq i, (i,j) \in E^\nu \\ 0 & j \neq i, (i,j) \notin E^\nu \\ -\sum_{k \in \mathcal{N}^\nu(i)} q_{ik}^\nu & j = i \end{cases}$$

**Departure intensity from GTFS:**

$$\mu_i^\nu = f_i^\nu \quad [\text{s}^{-1}]$$

$f_i^\nu$  [veh s<sup>-1</sup>]: peak-hour vehicle frequency at stop  $i$ ;  $\bar{k}^\nu$  [pax veh<sup>-1</sup>]: mean vehicle capacity;  $\rho^\nu \in (0, 1]$ : load factor (default 0.5).

# Kolmogorov Forward Equation (KFE)

## Theorem (KFE)

Let  $p^\vee(t)$  be the probability distribution of a tagged passenger. Under the Poisson-arrivals assumption:

$$\frac{dp^\vee(t)}{dt} = p^\vee(t) Q^\vee, \quad p^\vee(0) = p_0^\vee$$

Solution:  $p^\vee(t) = p_0^\vee \exp(Q^\vee t)$

**Conservation:**  $\frac{d}{dt} \|N^\vee(t)\|_1 = N^\vee(t) Q^\vee \mathbf{1} = 0$  (zero row sums of  $Q^\vee$ ). ✓

**Stationary distribution:**  $\pi^\vee Q^\vee = 0$ ,  $\|\pi^\vee\|_1 = 1$  (ergodic if  $G^\vee$  strongly connected).

## DTMC–CTMC link

$P^\vee(\Delta t) = \exp(Q^\vee \Delta t)$  is stochastic for all  $\Delta t \geq 0$ .

# Sparse Matrix Illustration

## Dense representation:

- $n \times n \approx 30$  M entries
- $\sim 230$  MB per mode
- 4 modes:  $\sim 920$  MB

**✗ infeasible**

## CSR (Compressed Sparse Row):

- $R = 22,656$  nonzeros total
- $\approx 0.6$  MB per mode
- 4 modes:  $\approx 2.4$  MB

**✓ mandatory**

CSR layout: `data[]`, `col_idx[]`, `row_ptr[]` — matrix–vector product in  $O(R)$ .

# Reaction-Channel Representation

Each off-diagonal generator entry is a *reaction channel*:

$$\mathcal{R}_{ij}^v : s_i \xrightarrow{q_{ij}^v} s_j, \quad (i,j) \in E^v$$

**Propensity** (effective rate for population  $N_i^v$ ):

$$a_{ij}^v(t) = N_i^v(t) \cdot q_{ij}^v$$

Total:  $a_0 = \sum_{v,i,j} a_{ij}^v$

## Gillespie Direct Method (exact)

- 1 Draw waiting time:  $\tau \sim \text{Exp}(a_0)$
- 2 Select channel  $(v^*, i^*, j^*)$  with probability  $\propto a_{ij}^v$  (inverse-CDF,  $O(R)$ )
- 3 Update:  $N_{i^*}^{v^*} - 1, N_{j^*}^{v^*} + 1, t \leftarrow t + \tau$

Produces exact sample paths of the CTMC.

# $\tau$ -Leaping for Large Populations

**Problem:** At BKK scale,  $a_0 \approx 4 \times 10^6 \text{ s}^{-1} \Rightarrow a_0 = \sum_{v,i} N_i^v f_i^v$ ; cost is feed/population dependent. **Not feasible.**

**Solution:**  $\tau$ -leaping [3]

$$\Delta k_{ij}^v \sim \text{Poisson}(a_{ij}^v \tau), \quad N_i^v \leftarrow N_i^v - \sum_j \Delta k_{ij}^v, \quad N_j^v \leftarrow N_j^v + \Delta k_{ij}^v$$

All channels fired in a *single vectorised NumPy call*.

## Advantages:

- $O(R)$  per step (vectorised)
- $\approx 2 \times 10^7$  ops for 900-s horizon
- Preserves stochastic fluctuations

## Caveats:

- Approximate (not exact)
- Oversubscribed stops use integer multinomial allocation:  
 $\sum_j \Delta k = N_i^v$
- Bounded propensity-ratio step heuristic (not full CGP)

*Hybrid:* large- $N$  stops  $\rightarrow$  fluid ODE; small- $N$  stops  $\rightarrow$  exact SSA.

# Prior Estimation of $N_i(0)$ : The Problem

## Available:

- GTFS timetable (public, CC0)
- Census data (KSH)
- OpenStreetMap (Overpass)
- BKK aggregate ridership  
( $N_{\text{day}} \approx 4 \times 10^6$ )

## Not available:

- AFC farecard tap-in data
- APC passenger counts
- OD survey data

## Critical limitation

Without validation data, model parameters  $\lambda^v$ ,  $\rho^v$ , and prior coefficients  $\beta$  are *assumed*, not estimated. Results are illustrative.

# Poisson Log-Linear Prior (E2 Estimator)

## Model:

$$N_i(0) \sim \text{Poisson}(\lambda_i), \quad \log \lambda_i = \beta_0 + \beta^\top \mathbf{x}_i + \xi_i$$

| Feature                              | Source         | $\hat{\beta}_k$ |
|--------------------------------------|----------------|-----------------|
| $\log(1 + f_i)$ : peak frequency     | GTFS           | +0.70           |
| $\log(1 + \text{Pop}_i)$ : pop.      | KSH/WorldPop   | +0.60           |
| $\log(1 + \text{POI}_i)$ : POI       | OSM            | +0.20           |
| $B_i$ : betweenness centrality       | GTFS graph     | +0.30           |
| $I_i$ : interchange indicator        | GTFS transfers | +0.50           |
| $\log(1 + D_i)$ : distance to centre | Haversine      | -0.25           |

**Calibration anchor** ( $\beta_0$  fixed by):

$$\sum_i \hat{N}_i(0) = N_{\text{peak}} = \underbrace{0.11}_{\phi_{\text{peak}}} \times 4 \times 10^6 \approx 440,000$$

# Computational Complexity Analysis

| Operation                    | Complexity                | Actual cost (BKK)         |
|------------------------------|---------------------------|---------------------------|
| Build sparse $Q^\vee$        | $O(R)$                    | trivial                   |
| KFE (LSODA, 900 s)           | $O(RT_{\text{steps}})$    | $\sim 2 \times 10^7$ ops  |
| Matrix exp (Krylov)          | $O(R\ell)$ , $\ell = 30$  | $\sim 7 \times 10^5$ /pt  |
| Exact SSA (15 min)           | $O(a_0 T)$                | feed/population dependent |
| $\tau$ -leap ( $\tau = 1$ s) | $O(R \cdot T/\tau)$       | $\sim 2 \times 10^7$ ops  |
| Resilience (ARPACK)          | sparse truncated spectrum | implementation dependent  |

## Practical recommendation

Use  $\tau$ -leaping for full-network, full-day simulation. Exact SSA only for sub-networks ( $\lesssim 500$  stops) or short windows ( $\lesssim 5$  min).

**Memory (CSR vs. dense):** Dense  $Q^\vee$ :  $\sim 230$  MB each. CSR:  $\sim 0.6$  MB each. Dense is completely infeasible.

# Top-10 Stops by Prior Estimate

| Rk | Stop                | $f_i$ (trips/h) | $\hat{N}_i(0)$ | $l_i$ |
|----|---------------------|-----------------|----------------|-------|
| 1  | Keleti pályaudvar M | 318             | 278            | ✓     |
| 2  | Blaha Lujza tér M   | 260             | 265            | ✓     |
| 3  | Széna tér           | 259             | 250            | ✓     |
| 4  | Margit híd, budai H | 277             | 238            | ✓     |
| 5  | Széll Kálmán tér M  | 227             | 228            | ✓     |
| 6  | Astoria M           | 215             | 227            | ✓     |

All top stops are interchange nodes ( $l_i = 1$ ). Platform duplicates (separate `stop_id` per direction) inflate some counts; consolidate by `parent_station`.

## E2 estimator

With full feature matrix anchored to  $N_{\text{peak}} = 440,000$ , spatial distribution unchanged; only overall scale shifts.

# KFE Dynamics: Mean-Field Regime

| Stop       | $t = 0$ | $t = 900 \text{ s}$ |
|------------|---------|---------------------|
| Keleti M   | 278     | 6.4                 |
| Blaha M    | 265     | 6.4                 |
| Széna tér  | 250     | 8.9                 |
| Sz. Kálmán | 228     | 8.9                 |
| Top-50 sum | 10 467  | 570                 |

Conservation:  $\Delta \|N\|_1 = 1.2 \times 10^{-9} \checkmark$

**Withdrawn legacy observation:** the former fast drain used pax/s as a per-passenger hazard and is not a corrected result.

**Correction:**  $\mu_i = f_i \text{ [s}^{-1}\text{]}$ ; the former pax/s rate was invalid.  $\Rightarrow$  drain time  $\approx 270/11 \approx 25 \text{ s}$ .

**Interpretation:** KFE = “frictionless” mean-field limit. It identifies the gravity-theoretic equilibrium  $\pi^\vee$ , *not* the operational trajectory.

# SSA vs. KFE: Regime Comparison

## Withdrawn table

The legacy comparison mixed incompatible modal stop indices and treated event-capped paths as completed horizons. It is retained only as an audit trail.

| Stop           | SSA <sub>mean</sub> | SSA <sub>05</sub> | SSA <sub>95</sub> | KFE | w (%) |
|----------------|---------------------|-------------------|-------------------|-----|-------|
| Mechwart liget | 269                 | 249               | 292               | 8.9 | 15.8  |
| Széna tér      | 255                 | 233               | 280               | 8.9 | 18.5  |
| Keleti M       | 145                 | 128               | 159               | 6.4 | 21.6  |

## Why the legacy comparison is invalid

- Event-capped SSA paths are right-censored and cannot represent  $T$ .
- Modal local indices cannot be summed without a global stop mapping.

# Criticality Framework

For each candidate stop  $i^*$ , compute node-removal metrics:

## Importance:

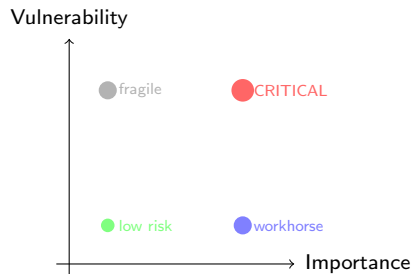
$$I_i = \sum_v N_{\text{tot}}^v \pi_i^v \mu_i^v \quad [\text{pax s}^{-1}]$$

## Vulnerability (three metrics):

- $\Delta K$ : Kemeny constant increase
- $\Delta g/g$ : spectral-gap collapse
- $\Delta E/E$ : network-efficiency loss

## Criticality:

$$C_i = \sqrt{\tilde{I}_i \tilde{V}_i}$$



# Resilience Results: v2.2 (Metro Aggregation Fix)

## Withdrawn ranking

The legacy eigenvalue calculation discarded sign/phase and applied Kemeny to reducible chains. No substantive top-15 claim is retained.

## Update from v2.1 $\rightarrow$ v2.2

Direction-specific platform stop-IDs collapsed to `parent_station`: metro platform aggregation is now tested; the displayed ranking is superseded.

| Rk | Stop (all Metro, mode 1)   | $\Delta K$ | $\Delta g/g$ | $C_i$        |
|----|----------------------------|------------|--------------|--------------|
| 1  | Deák Ferenc tér (M1+M2+M3) | 5735       | 1.00         | <b>1.000</b> |
| 2  | Kálvin tér (M3+M4)         | 3848       | 1.00         | 0.556        |
| 3  | Keleti pályaudvar (M2+M4)  | 1999       | 1.00         | 0.354        |
| 4  | Fővám tér                  | 1911       | 1.00         | 0.177        |
| 5  | Móricz Zsigmond körtér     | 1939       | 1.00         | 0.176        |
| 6  | Göncz Árpád városk.        | 1881       | 1.00         | 0.176        |
| 7  | Lehel tér                  | 1874       | 1.00         | 0.174        |

# Limitations: What the Model Cannot Do

## Most critical: no validation data

Without AFC/APC data, all quantitative results are *illustrative*, not operational.  $\lambda^v$ ,  $\rho^v$ , and  $\beta$  are assumed, not estimated.

| Limitation              | Path to remediation                      |
|-------------------------|------------------------------------------|
| Time-homogeneous rates  | Piecewise-constant rates; GTFS-RT        |
| No capacity constraints | Nonlinear $\mu_i(N_i)$                   |
| Single-leg transitions  | Trip-chaining state space $(i, v, \ell)$ |
| Origin-constrained only | Doubly-constrained IPF balancing         |
| Independent passengers  | Semi-Markov or agent-based model         |
| Exponential headways    | Phase-type / scheduled headway dist.     |

# Contributions Summary

- C1.** Rigorous CTMC generator from entropy maximisation, with dimensional consistency and sparse graph handling.
- C2.** DTMC–CTMC correspondence clarified: DTMC is the embedded jump chain, not uniformisation.
- C3.** Canonical global coordinates and completed-horizon metadata make KFE–SSA comparisons well-defined.
- C4.** Hierarchical demand prior (E1/E2/E3) anchored to BKK aggregate ridership.
- C5.** Reducibility-aware node removal with signed/complex eigenvalues retained for Kemeny calculations.

## Verification status

Legacy rankings are withdrawn. The bundled-feed smoke test builds 5,177 active nodes and conserves all 440,000 allocated passengers; new rankings require a pinned-feed full rerun.

# Future Work

- 1 **AFC calibration:** estimate  $\lambda^v$  from tap-in/tap-out distributions when farecard data are available.
- 2 **GTFS-RT integration:** update  $f_i^v$  in real time from vehicle-position feeds for time-adaptive CTMC.
- 3 **Doubly-constrained formulation:** IPF balancing for trip distribution.
- 4 **Capacity-dampened SSA:** nonlinear propensities with finite vehicle capacity.
- 5 **M5 metro scenario:** quantify reduction in  $\Delta K_{\text{Deak}}$  from a second north-south spine.
- 6 **Cross-city comparison:** apply to Vienna (WL), Prague (DPP), Warsaw (ZTM) GTFS feeds.
- 7 **Calendar-dates fix:** implement proper `calendar_dates.txt` weekday inference for metro service IDs (§10.3.4).

Code and data: [https://github.com/\[repo\]](https://github.com/[repo]) (MIT licence)

GTFS: <https://opendata.bkk.hu> (CC0-1.0)

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